LANE DEPARTURE WARNING SYSTEM (LDWS)

id092200029200

Outline

- The lane departure warning system (LDWS) recognizes vehicle lane lines on a road using the forward sensing camera (FSC) installed to the windshield, and if the vehicle departs from its lane unbeknownst to the driver, the system alerts the driver of the lane departure using a warning indication and warning sound.

Warning

- **The lane departure warning system (LDWS) is only a supplementary system for the prevention of a lane departure, however the lane departure warning system (LDWS) functions have limitations. Relying solely on the lane departure warning system (LDWS) and neglecting prudent steering wheel operation could cause an unexpected accident resulting in death or serious injury. Do not rely solely on the lane departure warning system (LDWS). Always make lane corrections using the steering wheel and drive safely.**
- **The lane departure warning system (LDWS) may not operate normally under the following conditions:**
  - **— Poor vehicle lane line visibility due to adverse weather (rain, snow, fog)**
  - **— Poor vehicle lane line visibility due to backlight, snow on the road, water puddles, road ruts**
  - **— Poor vehicle lane line visibility due to cracks and fissures in the vehicle lane**
  - **— Vehicle lane lines intersect with complexity due to additions or reductions of travel lanes**
  - **— Vehicle lane marking is not completely erased after road maintenance, two or more lane lines next to each other**
  - **— Vehicle lane lines not recognized due to vehicle ahead**

Caution

- **If the forward sensing camera (FSC) aiming is not completed correctly, the forward sensing camera (FSC) cannot record the camera shooting angle and the lane departure warning system (LDWS) may not operate normally. When performing the following servicing, always perform the forward sensing camera (FSC) aiming. For forward sensing camera (FSC) aiming procedure, refer to the Workshop Manual [FORWARD SENSING CAMERA (FSC) AIMING].**
  - **— Forward sensing camera (FSC) replacement**
  - **— Forward sensing camera (FSC) clip replacement**
  - **— Windshield replacement**

Function

- The forward sensing camera (FSC) sets warning-trigger lines based on the vehicle lane lines recognized by the camera, and if the vehicle contacts one of these warning-trigger lines it activates the warning indicator and warning sound. For details on the forward sensing camera (FSC) and warning-trigger lines, refer to [FORWARD SENSING CAMERA (FSC)]. (See FORWARD SENSING CAMERA (FSC).)

Driver operation determination function

- The forward sensing camera (FSC) determines the driver's intentions according to the following driving operations, and controls the warning operation.
  - — After the turn switch is operated, the vehicle lane is changed **within 60 s**.
  - — The steering wheel is turned at a certain angle or more, and the vehicle lane is changed.
  - — The accelerator pedal is depressed a certain amount or more, and the vehicle lane is changed.
  - — The brake pedal is depressed a certain amount or more and the vehicle departs from its lane.

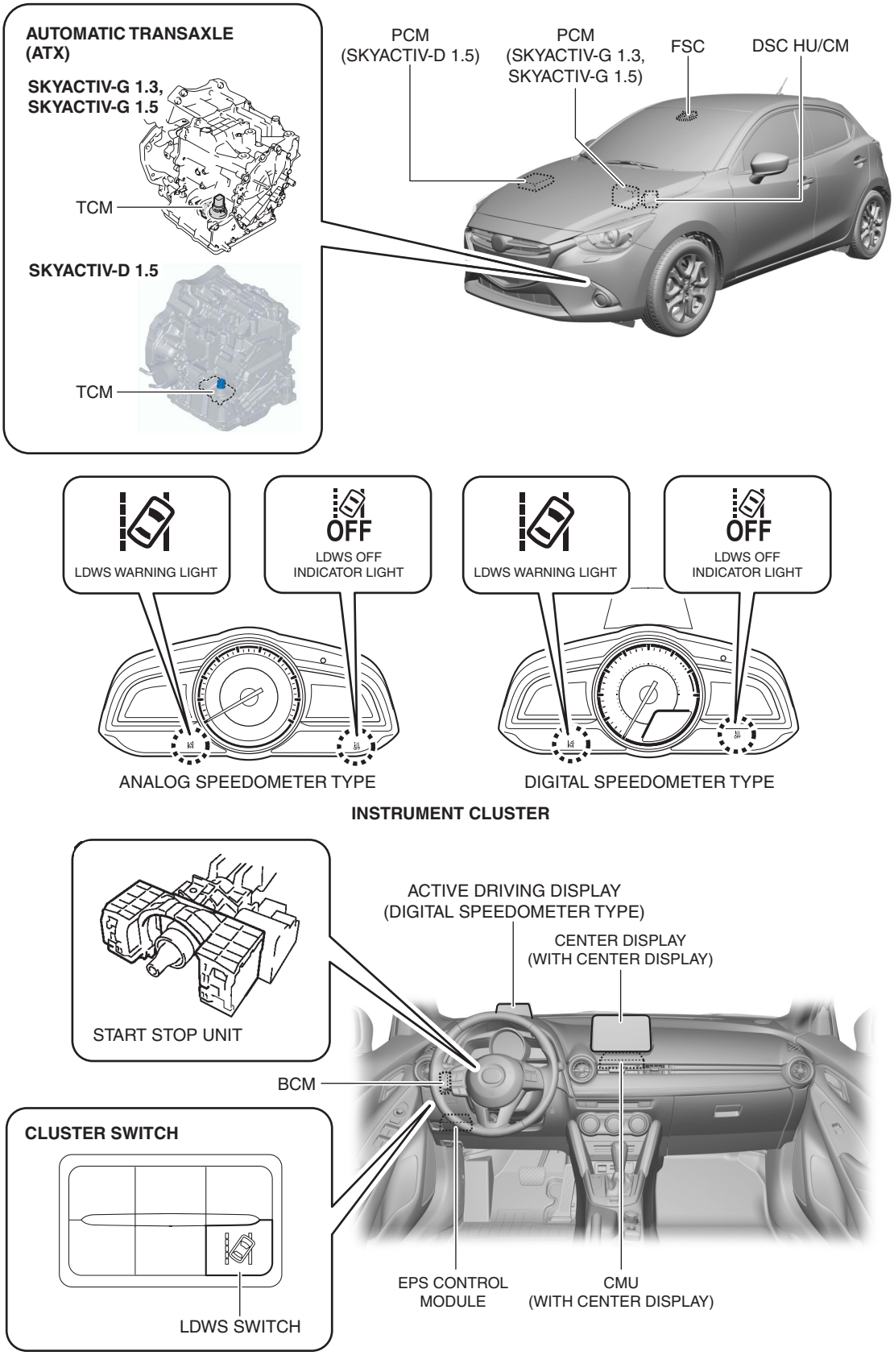
System conditions display function

- The forward sensing camera (FSC) displays the system status using the LDWS OFF indicator light and LDWS warning light in the instrument cluster, active driving display (digital speedometer type), and the warning display of the center display (with center display). For the warning display content of the center display, refer to [CENTER DISPLAY]. (See CENTER DISPLAY [WITH CENTER DISPLAY].)

LDWS switch	Condition	LDWS OFF indicator light	LDWS warning light	Active driving display (Digital speedometer type)	Center display (With center display)
OFF	System is off	On	Off	No display	No display

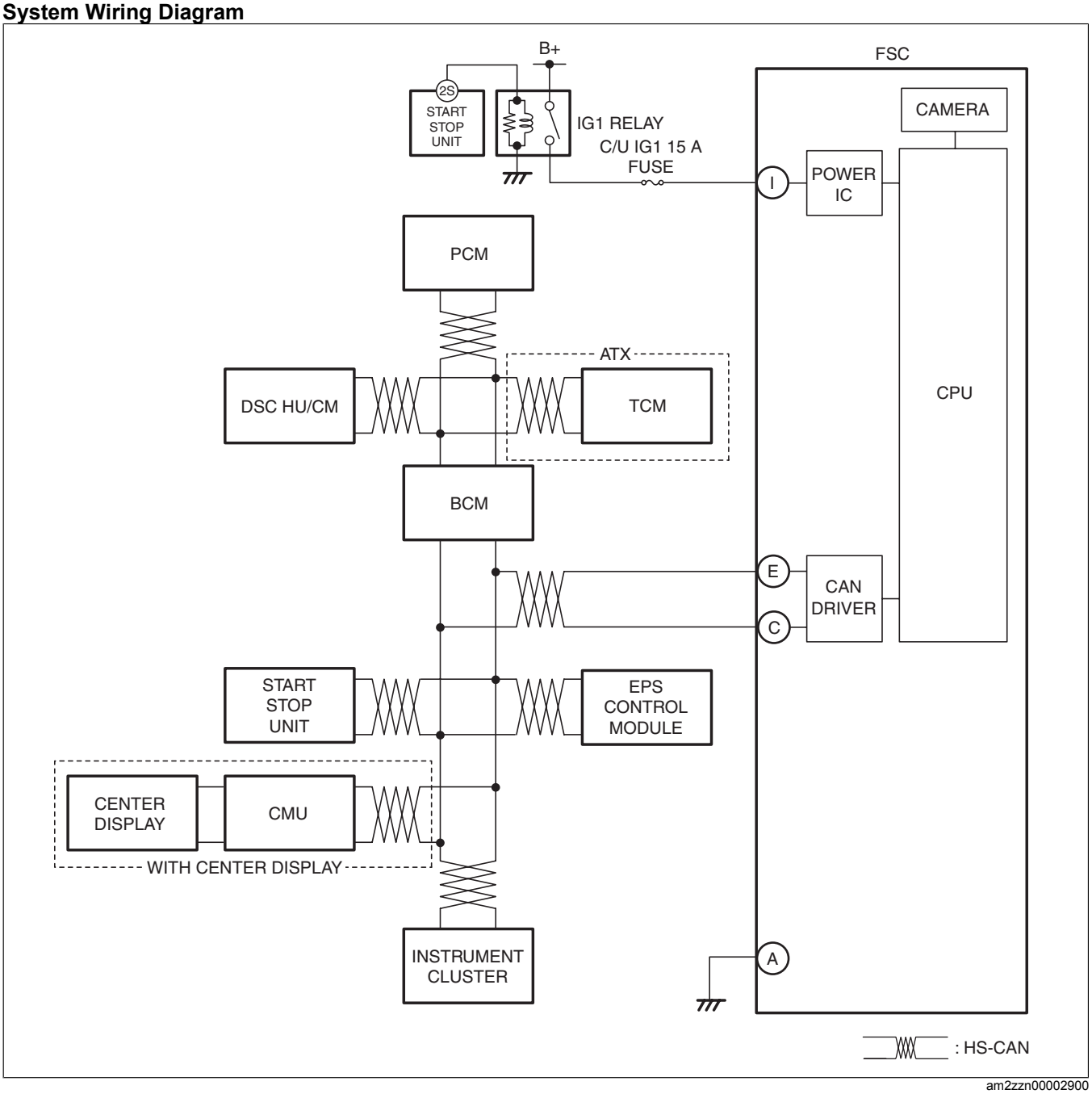
LDWS switch	Condition	LDWS OFF indicator light	LDWS warning light	Active driving display (Digital speedometer type)	Center display (With center display)
ON	Vehicle lane lines are recognized at a vehicle speed of **65 km/h {40 mph} or more**	Off	Off		No display
	While warning is triggered		Flash	 OR 	No display
	Vehicle speed is **60 km/h {37 mph} or less**		Off		No display
	Vehicle lane lines are not recognized				No display
	Forward sensing camera (FSC) detects camera fogging		On	No display	Warning display
	Forward sensing camera (FSC) detects that the camera is dirty			No display	Warning display
	Lane departure warning system (LDWS) has a malfunction			No display	Warning display
	Forward sensing camera (FSC) has a malfunction			No display	Warning display

Structural View



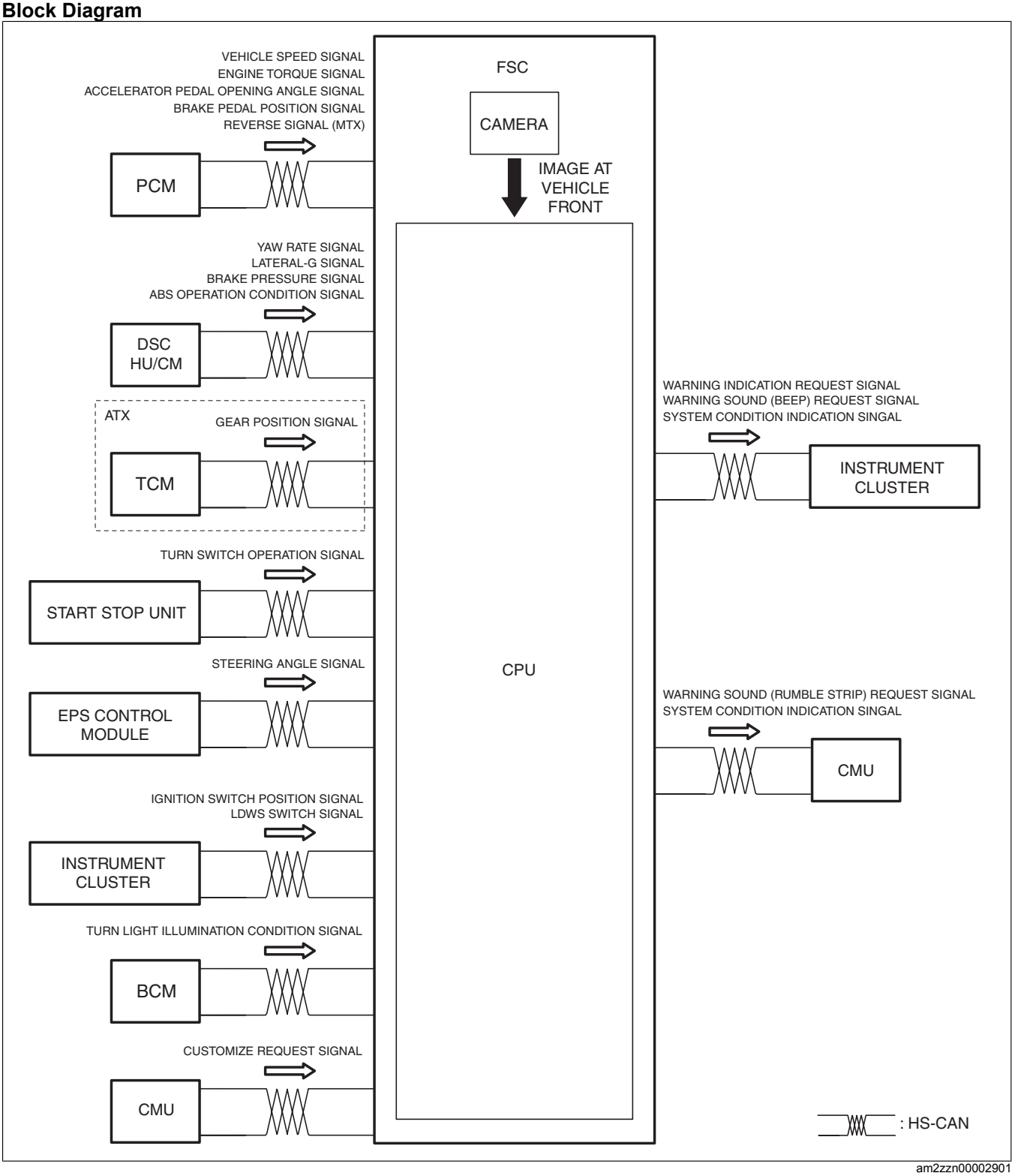
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System Wiring Diagram



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Block Diagram



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Operation

Operation conditions

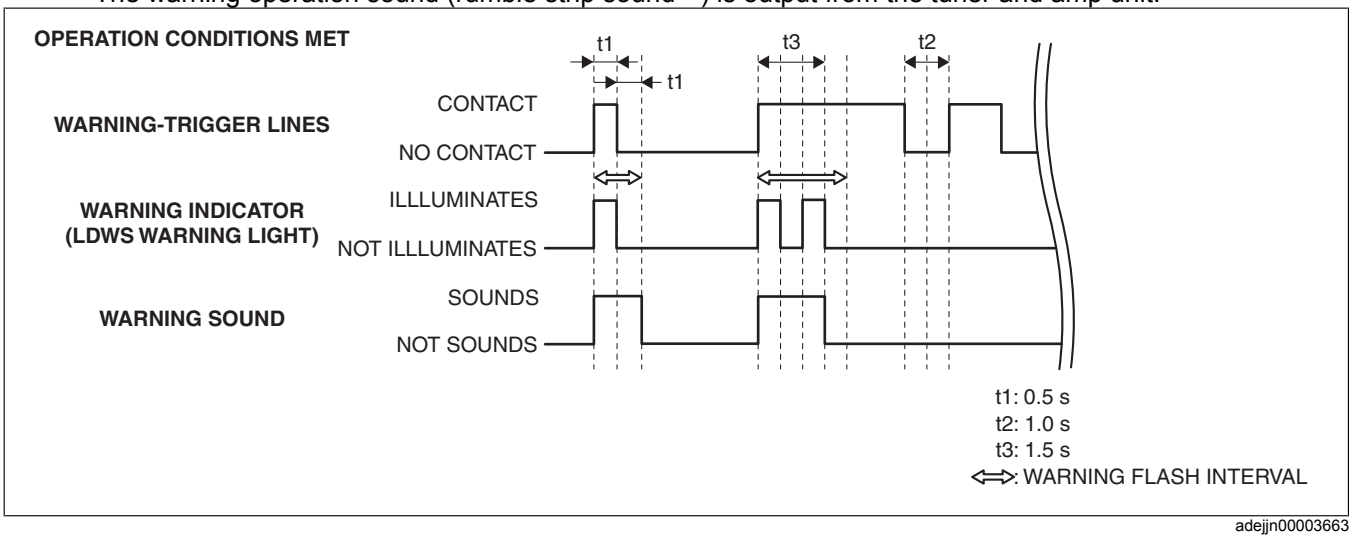
- The warning is triggered when all of the following conditions are met:
  - — LDWS switch is on (system on)
  - — Vehicle speed signal of **65 km/h {40 mph} or more** is received from PCM
  - — Vehicle contacts warning-trigger line unbeknownst to driver

Operation

- The period of time in which the warning is triggered differs by the period of time in which the vehicle is contacting the warning-trigger line.

— Vehicle contacts warning-trigger line for **less than 1 s**: Warning triggers for **1 s**
— Vehicle contacts warning-trigger line for **1 s or more**: Warning triggers for **1.5 s**

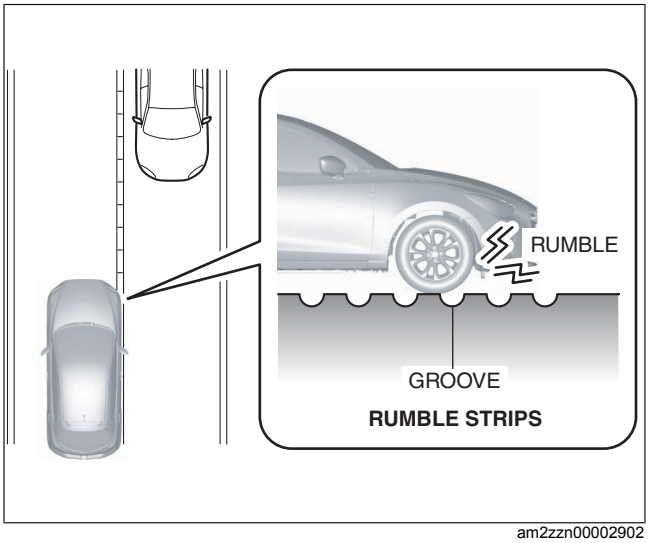
- If the warning is triggered one time, the warning will not trigger again until the vehicle travels within the warning-trigger lines for a period of **2 s**.
- The warning operation sound has the following two types:
  — The warning operation sound (beep) is output from the instrument cluster.
  — The warning operation sound (rumble strip sound*1) is output from the tuner and amp unit.



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*1 : A rumble strip is a series of grooves in the road pavement surface positioned at specific intervals, and when the vehicle passes over it a rumble sound is produced and an uncomfortable vibration can be felt which alerts the driver that the vehicle is departing from the lane lines. The rumble strip sound is a reproduction of the sound which occurs when a vehicle passes over a rumble strip.

Rumble strip sound

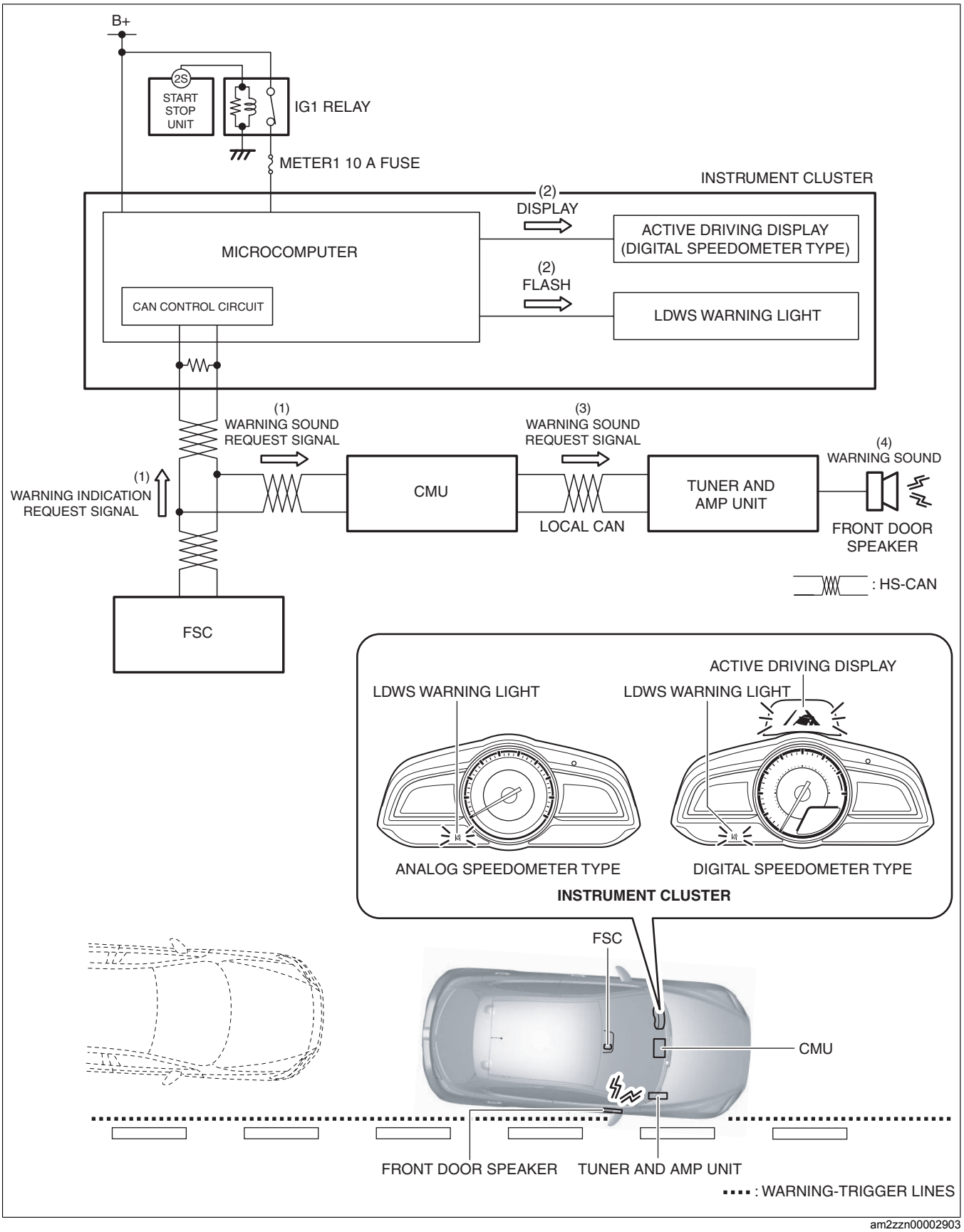


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1. If the forward sensing camera (FSC) determines that the vehicle has contacted a warning-trigger line, it sends (1) a warning indication request signal to the instrument cluster and a warning sound request signal to the connectivity master unit (CMU).
2. When the instrument cluster receives the warning indication request signal from the forward sensing camera (FSC), it flashes (2) the LDWS warning light and displays the warning screen in the active driving display (digital speedometer type).
3. When the connectivity master unit (CMU) receives the warning sound request signal from the forward sensing camera (FSC), it sends (3) the warning sound request signal to the tuner and amp unit.
4. When the tuner and amp unit receives the warning sound request signal from the connectivity master unit (CMU), it stops sound output from either the front left or right door speaker according to the request signal and outputs (4) a rumble strip sound.

Note

- If the user is conversing using the HF/TEL, or the voice recognition function is outputting voice, the rumble strip sound is not output.



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Beep sound

- For the operation, refer to [LDWS WARNING ALARM]. (See LDWS WARNING ALARM.)